A Practical Implementation of a Virtualized Energy Harvesting IoT

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I. The Battery Problem

Sensor networks monitor the environment in many applications, but battery-powered deployments need periodic battery replacement, which hinders scalable long-term operation.

Most deployed sensor nodes run on batteries or a grid connection. Batteries degrade, and replacing them can be costly or dangerous [1]; nodes are often discarded once their batteries are depleted, even when the hardware still works.

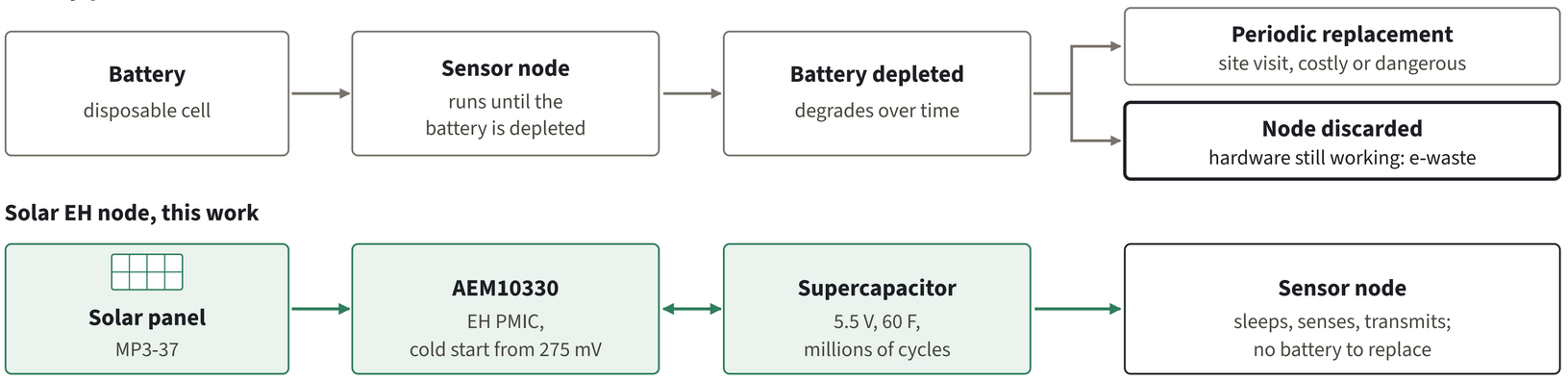
As IoT deployments grow, so does the need for batteries, and periodic battery maintenance limits how far monitoring networks can scale and how sustainable they are.

Solar energy harvesting (EH) lets nodes recharge from ambient sources and extends network lifetime [2], but harvested energy is intermittent [3].

EH, node virtualization and dynamic task assignment have been explored separately; few works implement them together in hardware.

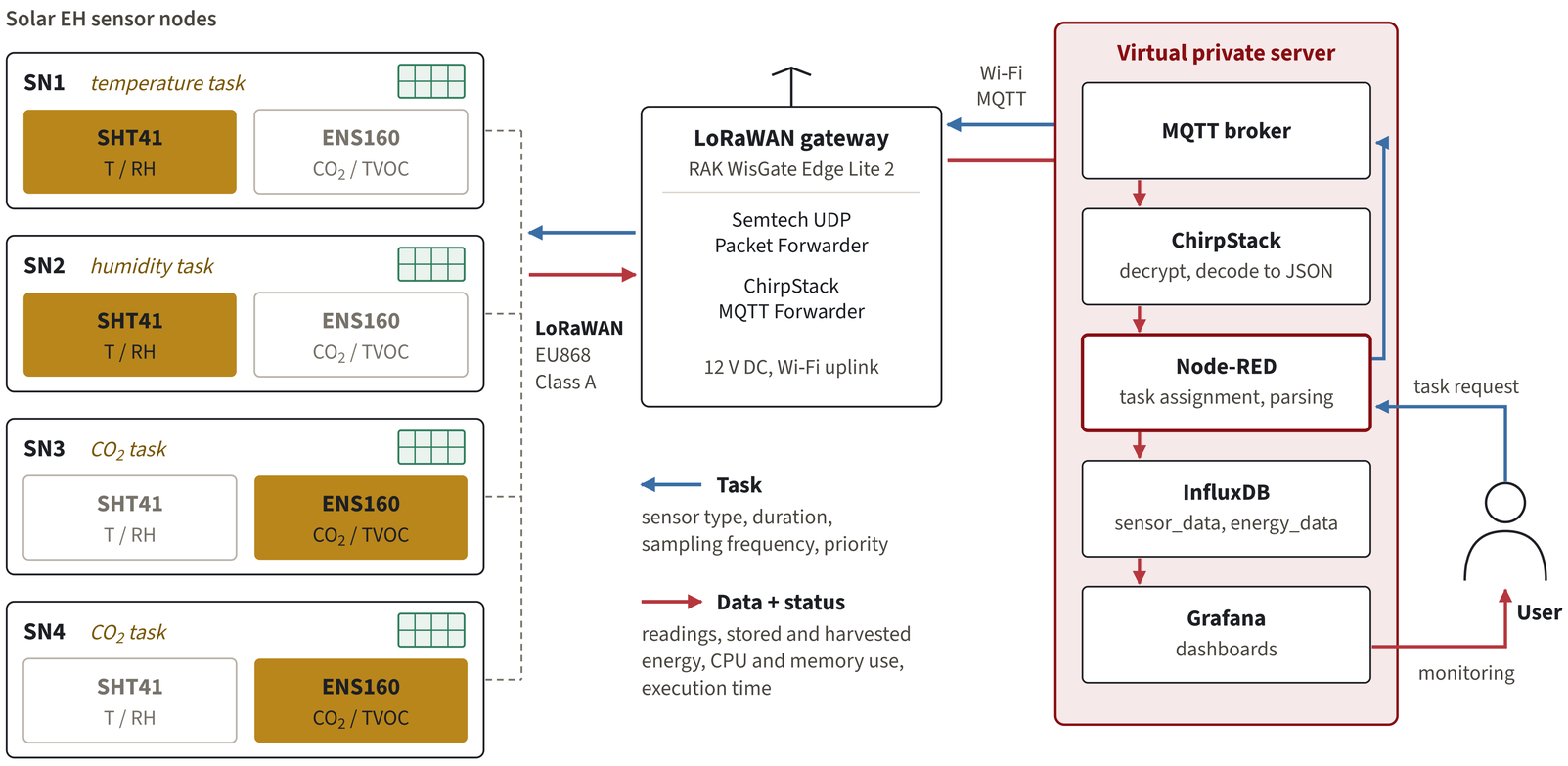
Goal: sustainable power without battery replacement, with every sensing task matched to the energy the nodes have stored.

II. Battery vs. Harvesting



	Battery or grid node	Solar EH node, this work
Energy	Disposable battery or a grid connection	Ambient sunlight through a solar panel
Storage	Rechargeable battery: thousands of cycles [4]	Supercapacitor: millions of cycles [4], wide temperature range [5]
Upkeep	Periodic battery replacement	No battery to replace, lower maintenance cost [6]
Hazardous sites	Workers visit industrial, remote and high-rise sites	Fewer site visits: lower operating cost, safer workers
End of life	Often discarded with working hardware	Less battery and electronic waste
Grid	Grid-connected nodes need a power line	Grid-free nodes; one gateway covers power- and cellular-limited sites

III. System Model

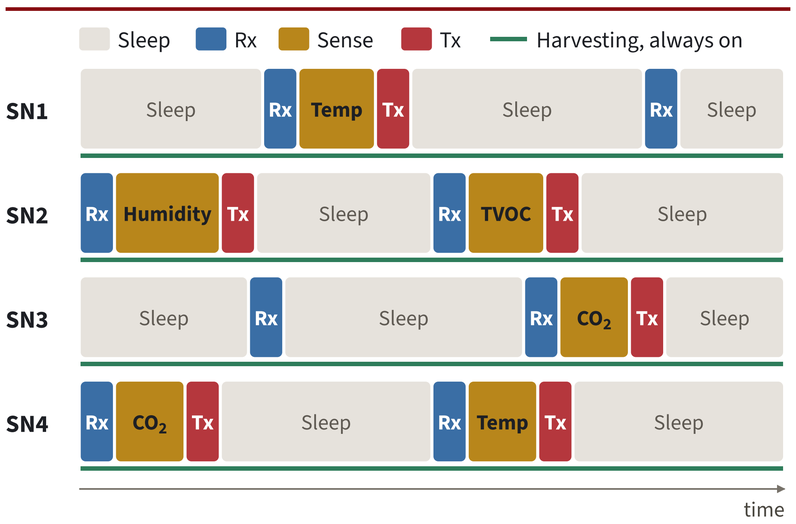


IV. Resilience for Sharjah

Theme	Where the network serves
Resilient energy systems	Solar nodes with no grid dependency keep sensing where power access is limited
Water security	Remote wastewater plants: hazardous gases, temperature and humidity; water management
Climate and environmental resilience	Greenhouses in UAE sustainable cities: humidity and CO_2
Smart and sustainable cities	Smart-city development such as Sharjah Sustainable City [7] and Masdar City [8]
Critical infrastructure	Industrial sites, remote infrastructure and high-rise installations, where access is difficult or unsafe

Energy-aware task assignment on battery-free nodes lays the groundwork for the large-scale adoption of sustainable IoT, enabling more resilient, efficient and environmentally conscious urban infrastructure.

V. Energy-Aware Node



Harvesting charges the supercapacitor whatever the MCU state. A node sleeps, wakes to listen for tasks, powers its sensors through a load switch only while a task runs, then transmits readings with its energy status.

Virtualization lets one node serve different sensing tasks over time, so each node is used efficiently [9].

VI. Energy-Aware Task Assignment

Each round, the VPS checks that the nodes' stored energy e_i covers every submitted task e_j plus a reserve $e_s = e_{\text{gas}} + e_{\text{temp}}$ kept for future immediate tasks:

$$\sum_{j \in \mathcal{T}_{\text{imm}} \cup \mathcal{T}_{\text{non}}} e_j + e_s \le \sum_{i=1}^{S} e_i \qquad (1)$$

If (1) fails, only immediate tasks enter. A MILP then favours costly tasks on nodes with a high remaining-energy share P_i, giving each task at most one node and keeping each node within its stored energy.

A two-phase heuristic is also defined: Phase 1 places each immediate task on the highest-energy node that can afford it; Phase 2 packs non-immediate tasks onto fewer nodes to reduce communication overhead, while each node keeps a reserve $\theta = e_{\text{gas}} + e_{\text{temp}}$. Unassigned tasks can be reconsidered once nodes report newly harvested energy.

Safety: a node must not start a task its stored energy cannot cover, and refuses high-power tasks when its supercapacitor runs low.

VII. Low-Power Design Choices

Part	Chosen	Over	Deciding figure
MCU	ATmega4809	ESP32	0.1 µA power-down [10] vs 10 µA deep sleep, RAM lost [11]
PMIC	AEM10330	BQ25570	Cold start from 275 mV [12] vs 600 mV [13]
Gas	ENS160	SGP30	29 mA vs 48.8 mA active [14], [15]
Radio	LoRaWAN	BLE, Zigbee	15 km line of sight vs 60 m and 75 m per hop [16]

VIII. Evaluation Plan

Requirement	Target
Node to dashboard delay	≤ 10 s
Data delivered, node to server	≥ 95 %
Average node power when idle	< 10 mW
Nodes served at once, no packet loss	≥ 4
Server and dashboard uptime	≥ 95 %

The PPK2 will profile every node mode to confirm sustainable operation during sunlight hours; the scheme will be judged on task completion rate, latency and network lifetime.

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